Problem Set 3

Dipankar Sahu, 727

11-02-2026

Problem to demonstrate the role of qualitative (nominal) predictors in addition to quantitative predictors in multiple linear regression.

Attach "Credits" data from R.

```
library(ISLR)

## Warning: package 'ISLR' was built under R version 4.4.3

data("Credit")
head(Credit)

##   ID  Income Limit Rating Cards Age Education Gender Student Married Ethni
city
## 1  1  14.891  3606    283     2  34        11   Male      No     Yes Cauca
sian
## 2  2 106.025  6645    483     3  82        15 Female     Yes     Yes     A
sian
## 3  3 104.593  7075    514     4  71        11   Male      No      No     A
sian
## 4  4 148.924  9504    681     3  36        11 Female      No      No     A
sian
## 5  5  55.882  4897    357     2  68        16   Male      No     Yes Cauca
sian
## 6  6  80.180  8047    569     4  77        10   Male      No      No Cauca
sian
##   Balance
## 1     333
## 2     903
## 3     580
## 4     964
## 5     331
## 6    1151

summary(Credit)

##        ID            Income           Limit           Rating
##  Min.   :  1.0   Min.   : 10.35   Min.   :  855   Min.   : 93.0
##  1st Qu.:100.8   1st Qu.: 21.01   1st Qu.: 3088   1st Qu.:247.2
##  Median :200.5   Median : 33.12   Median : 4622   Median :344.0
##  Mean   :200.5   Mean   : 45.22   Mean   : 4736   Mean   :354.9
##  3rd Qu.:300.2   3rd Qu.: 57.47   3rd Qu.: 5873   3rd Qu.:437.2
##  Max.   :400.0   Max.   :186.63   Max.   :13913   Max.   :982.0
##      Cards            Age          Education        Gender    Student
```

```
##  Min.   :1.000   Min.   :23.00   Min.   : 5.00    Male :193   No :360
##  1st Qu.:2.000   1st Qu.:41.75   1st Qu.:11.00   Female:207   Yes: 40
##  Median :3.000   Median :56.00   Median :14.00
##  Mean   :2.958   Mean   :55.67   Mean   :13.45
##  3rd Qu.:4.000   3rd Qu.:70.00   3rd Qu.:16.00
##  Max.   :9.000   Max.   :98.00   Max.   :20.00
##  Married              Ethnicity      Balance
##  No :155   African American: 99   Min.   :   0.00
##  Yes:245   Asian           :102   1st Qu.:  68.75
##            Caucasian       :199   Median : 459.50
##                                   Mean   : 520.01
##                                   3rd Qu.: 863.00
##                                   Max.   :1999.00
```

Regress “balance” on

(a) “gender” only.

```r
# Model (a): Balance ~ Gender
model_a <- lm(Balance ~ Gender, data = Credit)
summary(model_a)
```

```
##
## Call:
## lm(formula = Balance ~ Gender, data = Credit)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -529.54 -455.35  -60.17  334.71 1489.20
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    509.80      33.13  15.389   <2e-16 ***
## GenderFemale    19.73      46.05   0.429    0.669
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 460.2 on 398 degrees of freedom
## Multiple R-squared:  0.0004611,  Adjusted R-squared:  -0.00205
## F-statistic: 0.1836 on 1 and 398 DF,  p-value: 0.6685
```

(b) “gender” and “ethnicity” .

```r
# Model (b): Balance ~ Gender + Ethnicity
model_b <- lm(Balance ~ Gender + Ethnicity, data = Credit)
summary(model_b)
```

```
##
## Call:
## lm(formula = Balance ~ Gender + Ethnicity, data = Credit)
##
## Residuals:
```

```
##     Min      1Q  Median      3Q     Max
## -540.92 -453.61  -56.37  336.24 1490.77
##
## Coefficients:
##                     Estimate Std. Error t value Pr(>|t|)
## (Intercept)           520.88      51.90  10.036   <2e-16 ***
## GenderFemale           20.04      46.18   0.434    0.665
## EthnicityAsian        -19.37      65.11  -0.298    0.766
## EthnicityCaucasian    -12.65      56.74  -0.223    0.824
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 461.3 on 396 degrees of freedom
## Multiple R-squared:  0.000694,   Adjusted R-squared:  -0.006877
## F-statistic: 0.09167 on 3 and 396 DF,  p-value: 0.9646
```

(c) "gender", "ethnicity", "income".

```r
# Model (c): Balance ~ Gender + Ethnicity + Income
model_c <- lm(Balance ~ Gender + Ethnicity + Income, data = Credit)
summary(model_c)
```

```
##
## Call:
## lm(formula = Balance ~ Gender + Ethnicity + Income, data = Credit)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -794.14 -351.67  -52.02  328.02 1110.09
##
## Coefficients:
##                     Estimate Std. Error t value Pr(>|t|)
## (Intercept)         230.0291    53.8574   4.271 2.44e-05 ***
## GenderFemale         24.3396    40.9630   0.594    0.553
## EthnicityAsian        1.6372    57.7867   0.028    0.977
## EthnicityCaucasian    6.4469    50.3634   0.128    0.898
## Income                6.0542     0.5818  10.406  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 409.2 on 395 degrees of freedom
## Multiple R-squared:  0.2157, Adjusted R-squared:  0.2078
## F-statistic: 27.16 on 4 and 395 DF,  p-value: < 2.2e-16
```

(d) Output all the regressions in (a)-(c) in a single table using stargazer. Comment on the significant coefficients in each of the models.

```r
library(stargazer)
```

```
##
## Please cite as:
```

```
##  Hlavac, Marek (2022). stargazer: Well-Formatted Regression and Summary St
atistics Tables.

##  R package version 5.2.3. https://CRAN.R-project.org/package=stargazer

stargazer(model_a, model_b, model_c, type = "html", out = "f1.html")

##
## <table style="text-align:center"><tr><td colspan="4" style="border-bottom:
1px solid black"></td></tr><tr><td style="text-align:left"></td><td colspan="
3"><em>Dependent variable:</em></td></tr>
## <tr><td></td><td colspan="3" style="border-bottom: 1px solid black"></td><
/tr>
## <tr><td style="text-align:left"></td><td colspan="3">Balance</td></tr>
## <tr><td style="text-align:left"></td><td>(1)</td><td>(2)</td><td>(3)</td><
/tr>
## <tr><td colspan="4" style="border-bottom: 1px solid black"></td></tr><tr><
td style="text-align:left">GenderFemale</td><td>19.733</td><td>20.038</td><td
>24.340</td></tr>
## <tr><td style="text-align:left"></td><td>(46.051)</td><td>(46.178)</td><td
>(40.963)</td></tr>
## <tr><td style="text-align:left"></td><td></td><td></td><td></td></tr>
## <tr><td style="text-align:left">EthnicityAsian</td><td></td><td>-19.371</t
d><td>1.637</td></tr>
## <tr><td style="text-align:left"></td><td></td><td>(65.107)</td><td>(57.787
)</td></tr>
## <tr><td style="text-align:left"></td><td></td><td></td><td></td></tr>
## <tr><td style="text-align:left">EthnicityCaucasian</td><td></td><td>-12.65
3</td><td>6.447</td></tr>
## <tr><td style="text-align:left"></td><td></td><td>(56.740)</td><td>(50.363
)</td></tr>
## <tr><td style="text-align:left"></td><td></td><td></td><td></td></tr>
## <tr><td style="text-align:left">Income</td><td></td><td></td><td>6.054<sup
>***</sup></td></tr>
## <tr><td style="text-align:left"></td><td></td><td></td><td>(0.582)</td></t
r>
## <tr><td style="text-align:left"></td><td></td><td></td><td></td></tr>
## <tr><td style="text-align:left">Constant</td><td>509.803<sup>***</sup></td
><td>520.880<sup>***</sup></td><td>230.029<sup>***</sup></td></tr>
## <tr><td style="text-align:left"></td><td>(33.128)</td><td>(51.901)</td><td
>(53.857)</td></tr>
## <tr><td style="text-align:left"></td><td></td><td></td><td></td></tr>
## <tr><td colspan="4" style="border-bottom: 1px solid black"></td></tr><tr><
td style="text-align:left">Observations</td><td>400</td><td>400</td><td>400</
td></tr>
## <tr><td style="text-align:left">R<sup>2</sup></td><td>0.0005</td><td>0.001
</td><td>0.216</td></tr>
## <tr><td style="text-align:left">Adjusted R<sup>2</sup></td><td>-0.002</td>
<td>-0.007</td><td>0.208</td></tr>
## <tr><td style="text-align:left">Residual Std. Error</td><td>460.230 (df =
```

```
398)</td><td>461.337 (df = 396)</td><td>409.218 (df = 395)</td></tr>
## <tr><td style="text-align:left">F Statistic</td><td>0.184 (df = 1; 398)</t
d><td>0.092 (df = 3; 396)</td><td>27.161<sup>***</sup> (df = 4; 395)</td></tr
>
## <tr><td colspan="4" style="border-bottom: 1px solid black"></td></tr><tr><
td style="text-align:left"><em>Note:</em></td><td colspan="3" style="text-ali
gn:right"><sup>*</sup>p<0.1; <sup>**</sup>p<0.05; <sup>***</sup>p<0.01</td></
tr>
## </table>
```

Comments on Significant Coefficients:

Model (a) - Gender only:

- The intercept (509.803) represents the average balance for the reference group, which is males.
- The coefficient on GenderFemale (19.733) means females have, on average, about $19.73 higher balance than males, but this effect is not statistically significant (large standard error, no stars).
- R^2 = 0.0005, meaning gender explains virtually none of the variation in balance.

Model (b) - Gender and Ethnicity:

- The intercept (520.880) represents the average balance for the baseline group (Male & reference ethnicity).
- GenderFemale (20.038) remains statistically insignificant. EthnicityAsian (-19.371) and EthnicityCaucasian (-12.653) are also statistically insignificant.
- The model still has very low explanatory power (R^2 = 0.001).

Model (c) - Gender, Ethnicity, and Income:

- The intercept (230.029) represents the expected balance for the baseline group when income is zero.
- Income (6.054) is highly statistically significant. Interpretation: A one-unit increase in income is associated with an average increase of about $6.05 in balance, holding gender and ethnicity constant.
- Gender and ethnicity variables remain statistically insignificant.
- R^2 increases dramatically to 0.216, indicating that income explains about 21.6% of the variation in balance.

(e) Explain how gender affects “balance” in each of the models (a)- (c) .

```r
gender_coef_a <- coef(model_a)["GenderFemale"]
gender_coef_b <- coef(model_b)["GenderFemale"]
gender_coef_c <- coef(model_c)["GenderFemale"]

cat("Gender (Female) coefficient in Model (a):", round(gender_coef_a, 2), "\n
")
```

```
## Gender (Female) coefficient in Model (a): 19.73

cat("Gender (Female) coefficient in Model (b):", round(gender_coef_b, 2), "\n
")

## Gender (Female) coefficient in Model (b): 20.04

cat("Gender (Female) coefficient in Model (c):", round(gender_coef_c, 2), "\n
")

## Gender (Female) coefficient in Model (c): 24.34
```

Interpretation:

- **Model (a):** Females have, on average, $19.73 higher balance than males, but this difference is not statistically significant.
- **Model (b):** After controlling for ethnicity, females have $20.04 higher balance than males, and this difference remains statistically insignificant.
- **Model (c):** After controlling for ethnicity and income, females have $24.34 higher balance than males, but this difference is still not statistically significant.

Conclusion: Gender does not have a statistically significant effect on credit card balance in any of the three models, even after controlling for ethnicity and income.

(f) Compare the average credit card balance of a male African with a male Caucasian on the basis of model (b).

```
# Model (b) coefficients
coef_b <- coef(model_b)

balance_male_african <- coef_b["(Intercept)"]

balance_male_caucasian <- coef_b["(Intercept)"] + coef_b["EthnicityCaucasian"
]

cat("Average Balance for Male African American:",
    round(balance_male_african, 2), "\n")

## Average Balance for Male African American: 520.88

cat("Average Balance for Male Caucasian:",
    round(balance_male_caucasian, 2), "\n")

## Average Balance for Male Caucasian: 508.23

cat("Difference (Caucasian - African American):",
    round(balance_male_caucasian - balance_male_african, 2), "\n")

## Difference (Caucasian - African American): -12.65
```

Interpretation: Based on Model (b), a male Caucasian is predicted to have $12.65 lower credit card balance compared to a male African American, on average. However, this difference is not statistically significant.

(g) Compare the average credit card balance of a male African with a male Caucasian when each earns 100,000 dollars. For comparison, use the model in (c).

```
# Model (c) coefficients
coef_c <- coef(model_c)

balance_male_african_100k <- coef_c["(Intercept)"] +
                             coef_c["Income"] * 100

balance_male_caucasian_100k <- coef_c["(Intercept)"] +
                               coef_c["EthnicityCaucasian"] +
                               coef_c["Income"] * 100

cat("Average Balance for Male African American (Income = $100k):",
    round(balance_male_african_100k, 2), "\n")

## Average Balance for Male African American (Income = $100k): 835.45

cat("Average Balance for Male Caucasian (Income = $100k):",
    round(balance_male_caucasian_100k, 2), "\n")

## Average Balance for Male Caucasian (Income = $100k): 841.9

cat("Difference (Caucasian - African American):",
    round(balance_male_caucasian_100k - balance_male_african_100k, 2), "\n")

## Difference (Caucasian - African American): 6.45
```

Interpretation: When both individuals earn $100,000, a male Caucasian is predicted to have $6.45 higher balance compared to a male African American. The difference is almost entirely due to ethnicity, not income.

(h) Compare and comment on the answers in (f) and (g)

```
diff_f <- balance_male_caucasian - balance_male_african
diff_g <- balance_male_caucasian_100k - balance_male_african_100k

cat("Difference in (f):", round(diff_f, 2), "\n")

## Difference in (f): -12.65

cat("Difference in (g):", round(diff_g, 2), "\n")

## Difference in (g): 6.45

cat("Change in difference:", round(diff_g - diff_f, 2), "\n")

## Change in difference: 19.1
```

Comments:

1. **Similar Differences:** Both models predict very similar differences between male Caucasians and male African Americans (approximately −$12.65 to −$12.73), reflecting the ethnicity coefficient in each model.

2. **Income Effect:** The inclusion of income in Model (c) does not substantially change the ethnicity effect because the model is additive. Income shifts predicted balances for all groups equally, so the gap between ethnic groups remains essentially the ethnicity coefficient.

3. **Statistical Significance:** Neither ethnicity nor gender differences are statistically significant in either model, as indicated by the absence of significance stars and large standard errors relative to the coefficients.

4. **Income Dominates:** In Model (c), income is the only statistically significant predictor. However, it explains about 21.6% of the variation in balance ($R^2 = 0.216$), While income substantially improves explanatory power compared to Models (a) and (b), a large portion of the variation in balance remains unexplained.

(i) Based on the model in (c), predict the credit card balance of a female Asian whose income is 2000,000 dollars.

```
new_data <- data.frame(
  Gender = "Female",
  Ethnicity = "Asian",
  Income = 200
)

# Predict using model (c)
predicted_balance <- predict(model_c, newdata = new_data, interval = "confide
nce")

cat("Predicted Balance for Female Asian with $200,000 income:\n")

## Predicted Balance for Female Asian with $200,000 income:

cat("Point Estimate:", round(predicted_balance[1], 2), "\n")

## Point Estimate: 1466.85

cat("95% Confidence Interval: [", round(predicted_balance[2], 2), ",",
    round(predicted_balance[3], 2), "]\n")

## 95% Confidence Interval: [ 1267.81 , 1665.89 ]
```

Interpretation: A female Asian individual with an income of $200,000 is predicted to have a credit card balance of approximately $1466.85, with a 95% confidence interval of [1267.81, 1665.89].

(j) Check the goodness of fit of the different models in (a) -(c) in terms of adjusted R2. Which model would you prefer?

```
get_fit_stats <- function(model) {
  adj_r_squared <- summary(model)$adj.r.squared

  return(c(Adj_R2 = adj_r_squared))
}

fit_comparison <- data.frame(
  Model_a = get_fit_stats(model_a),
  Model_b = get_fit_stats(model_b),
  Model_c = get_fit_stats(model_c)
)

print(round(fit_comparison, 4))

##        Model_a Model_b Model_c
## Adj_R2 -0.0021 -0.0069  0.2078

fit_comparison_long <- data.frame(
  Metric = rownames(fit_comparison),
  Model_a = fit_comparison$Model_a,
  Model_b = fit_comparison$Model_b,
  Model_c = fit_comparison$Model_c
)

knitr::kable(
  fit_comparison_long,
  col.names = c("Metric", "Model (a)", "Model (b)", "Model (c)"),
  caption = "Model Comparison: Adjusted R-squared",
  digits = 4
)
```

Model Comparison: Adjusted R-squared

Metric	Model (a)	Model (b)	Model (c)
Adj_R2	-0.0021	-0.0069	0.2078

Model Selection Recommendation:

Model (c) is clearly the preferred model based on the following criteria:

1. **Adjusted R^2:** Model (c) has the highest adjusted R^2 (0.2078), explaining 20.78% of variance compared to essentially 0% for models (a) and (b).

Problem to demonstrate the impact of ignoring interaction term in multiple linear regression

```
set.seed(123)
```

```r
n <- 100  # sample size
R <- 1000  # number of repetitions

config1 <- c(beta0 = -2.5, beta1 = 1.2, beta2 = 2.3, beta3 = 0.001)
config2 <- c(beta0 = -2.5, beta1 = 1.2, beta2 = 2.3, beta3 = 3.1)
```

Consider a simulation setting where the data is generated as follows:

Step 1: Generate x_{1i} from Normal(0,1) distribution, i = 1, 2, .., n

Step 2: Generate x_{2i} from Bernoulli (0.3) distribution, i = 1, 2, .., n

Step 3: Generate ε_i from Normal(0,1) and hence generate the response $y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \beta_3 (x_{1i} \times x_{2i}) + \varepsilon_i$, i = 1, 2, , , n.

Step 4: Run two separate multiple linear regressions (i) using the model in Step 3 and (ii) using the model in Step 3 without the interaction term. Repeat Steps 1-4 , R = 1000 times. At each simulation compute the MSE for the correct model (i.e. model with the interaction term) and the naive model (i.e. the model without the interaction term). Finally find the average MSE's for each model. From the output, demonstrate the impact of ignoring the interaction term.

Carry out the analysis for n = 100 and the following parametric configurations: (β_0, β_1, β_2, β_3) = (−2.5, 1.2, 2.3, 0.001) , (-2.5, 1.2. 2.3, 3.1). Set seed as 123.

```r
set.seed(123)

n <- 100
R <- 1000

param_list <- list(
  c(-2.5, 1.2, 2.3, 0.001),
  c(-2.5, 1.2, 2.3, 3.1)
)

run_simulation <- function(beta) {

  beta0 <- beta[1]
  beta1 <- beta[2]
  beta2 <- beta[3]
  beta3 <- beta[4]

  mse_correct <- numeric(R)
  mse_naive   <- numeric(R)

  for (r in 1:R) {

    x1 <- rnorm(n, 0, 1)
    x2 <- rbinom(n, 1, 0.3)
```

```r
    eps <- rnorm(n, 0, 1)
    y <- beta0 + beta1*x1 + beta2*x2 + beta3*(x1*x2) + eps

    model_correct <- lm(y ~ x1 * x2)

    model_naive <- lm(y ~ x1 + x2)

    mse_correct[r] <- mean((y - predict(model_correct))^2)
    mse_naive[r]   <- mean((y - predict(model_naive))^2)
  }

  return(c(
    Avg_MSE_Correct = mean(mse_correct),
    Avg_MSE_Naive   = mean(mse_naive)
  ))
}

results_1 <- run_simulation(param_list[[1]])
results_2 <- run_simulation(param_list[[2]])

results <- rbind(
  "beta3 = 0.001 (≈ no interaction)" = results_1,
  "beta3 = 3.1 (strong interaction)" = results_2
)

round(results, 4)

##                                   Avg_MSE_Correct Avg_MSE_Naive
## beta3 = 0.001 (≈ no interaction)          0.9632        0.9739
## beta3 = 3.1 (strong interaction)          0.9578        2.8633
```

Interpretation: When the true interaction effect is essentially zero, omitting it barely hurts predictive accuracy. The naive additive model performs almost as well as the correctly specified model.

When the interaction truly matters, ignoring it leads to severe model misspecification and dramatically higher prediction error.